

Optimization

Minimization and Karnaugh Maps

- You can simplify, reduce the cost of, the circuit (i.e. logical expression) using two ways:

1. Boolean Algebra properties
2. K-map

- A Karnaugh Map (K-Map) is a visual method used in digital system design to simplify Boolean expressions, reducing hardware complexity and cost.

- It organizes truth table values into a grid.

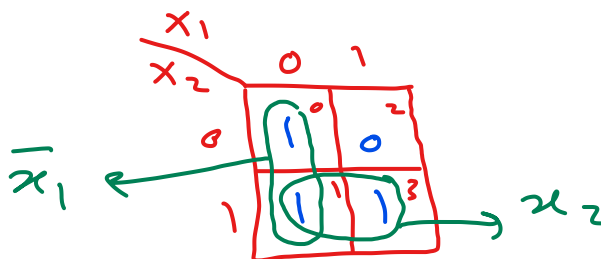
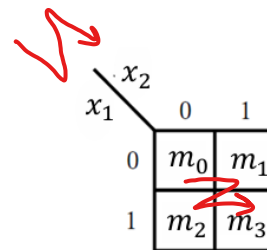
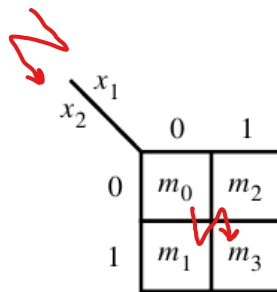
- A grid representation of the truth table, where adjacent cells differ by only one variable.

- Simplify (or minimize) SOP functions (circuits):

1- Two Variables Map: $2^2 = 4$

$$f(x_1, x_2) = \bar{x}_1 \bar{x}_2 + \bar{x}_1 x_2 + x_1 x_2$$

x_1	x_2	f
0	0	m_0 1
0	1	m_1 1
1	0	m_2 0
1	1	m_3 1



$$f(x_1, x_2) = \bar{x}_1 + x_2$$

2- Three Variables Map: $2^3 = 8$ cells

①

x_1	x_2	x_3	f
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

②

$x_1 x_2$	00	01	11	10
0	m_0	m_2	m_6	m_4
1	m_1	m_3	m_7	m_5

x_1	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

Example 1: Simplify the following using a K-map

$$f(x_1, x_2, x_3) = \overline{x_1} \overline{x_2} x_3 + x_1 \overline{x_2} x_3 + x_1 \overline{x_3}$$

SOP

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

$x_1 x_2$	00	01	11	10
0	0	0	1	1
1	1	0	0	1

$$f = x_1 \overline{x_3} + \overline{x_2} x_3 \quad \text{min cost sop}$$

Example 2: Give the minimum cost representation of the following logical expression

$$f(a,b,c) = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b} + ab\bar{c}$$

$\bar{a}$	b	$\bar{c}$	f
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

$\bar{a}b$				
c	00	01	11	10
0	1 ⁰	1 ²	1 ⁶	1 ⁴
1	0 ¹	0 ³	0 ⁷	1 ⁵

$\bar{c}$ (red circle around 00, 01, 11, 10 in row 0)

$a\bar{b}$ (green circle around 10 in row 1)

$$f = \bar{c} + a\bar{b}$$

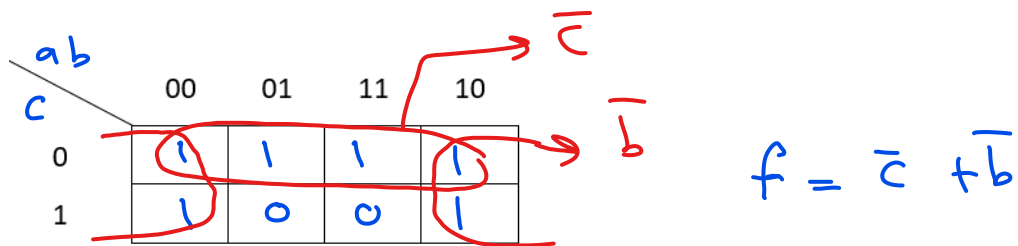
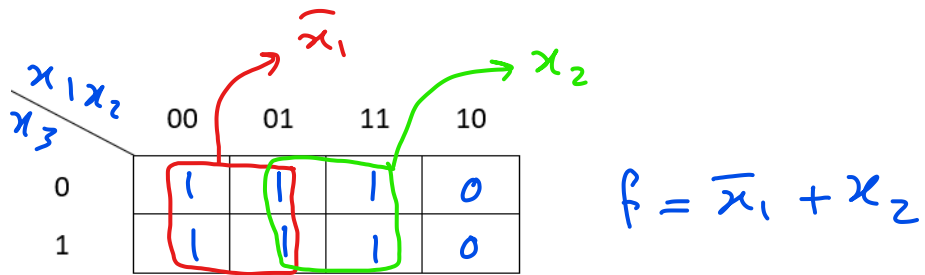
$\bar{a}b$				
c	00	01	11	10
0	1 ⁰	1 ²	0 ⁶	1 ⁴
1	1 ¹	1 ³	0 ⁷	1 ⁵

$\bar{a}$ (red circle around 00, 01, 10 in row 0)

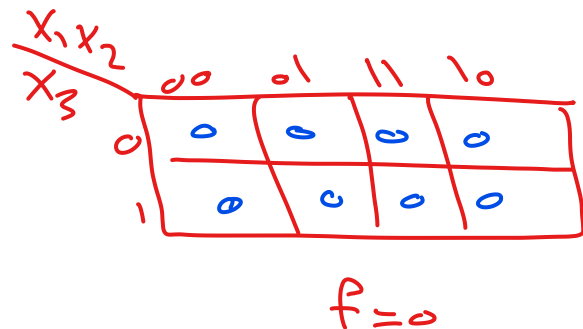
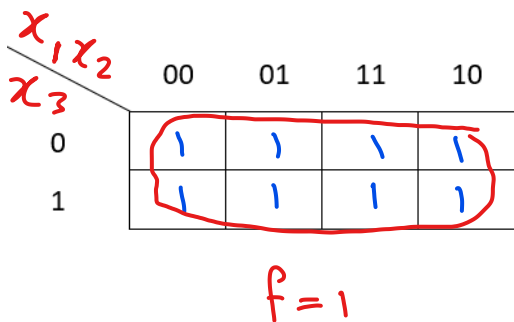
$\bar{b}$ (blue circle around 10 in row 0)

$$f = \bar{a} + \bar{b}b$$

Multiple examples:



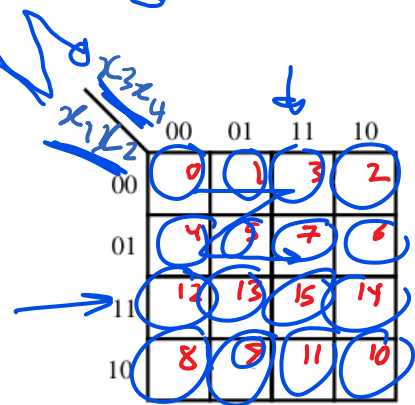
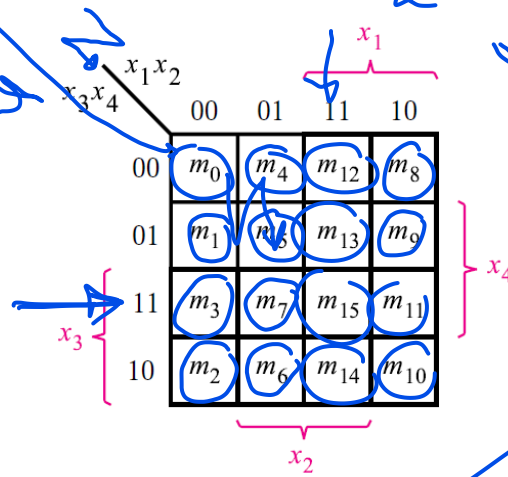
Special cases



3- Four Variables Map:

$$2^4 = 16 \text{ cells}$$

x1	x2	x3	x4	f
0	0	0	0	m_0
0	0	0	1	m_1
0	0	1	0	m_2
0	0	1	1	m_3
0	1	0	0	m_4
0	1	0	1	m_5
0	1	1	0	m_6
0	1	1	1	m_7
1	0	0	0	m_8
1	0	0	1	m_9
1	0	1	0	m_{10}
1	0	1	1	m_{11}
1	1	0	0	m_{12}
1	1	0	1	m_{13}
1	1	1	0	m_{14}
1	1	1	1	m_{15}



Example 1: Find the minimum cost SOP expression for the following function. (or Design the simplest SOP expression that implements the following expression)

$$f(x_1, x_2, x_3, x_4) = \sum m(\underline{2}, \underline{3}, \underline{9}, \underline{10}, 11, 13) \rightarrow \text{SOP}$$

	X1	X2	X3	X4	f
m ₀	0	0	0	0	0
m ₁	0	0	0	1	0
m ₂	0	0	1	0	1
m ₃	0	0	1	1	1
m ₄	0	1	0	0	0
m ₅	0	1	0	1	0
m ₆	0	1	1	0	0
m ₇	0	1	1	1	0
m ₈	1	0	0	0	0
m ₉	1	0	0	1	1
m ₁₀	1	0	1	0	1
m ₁₁	1	0	1	1	1
m ₁₂	1	1	0	0	0
m ₁₃	1	1	0	1	1
m ₁₄	1	1	1	0	0
m ₁₅	1	1	1	1	0

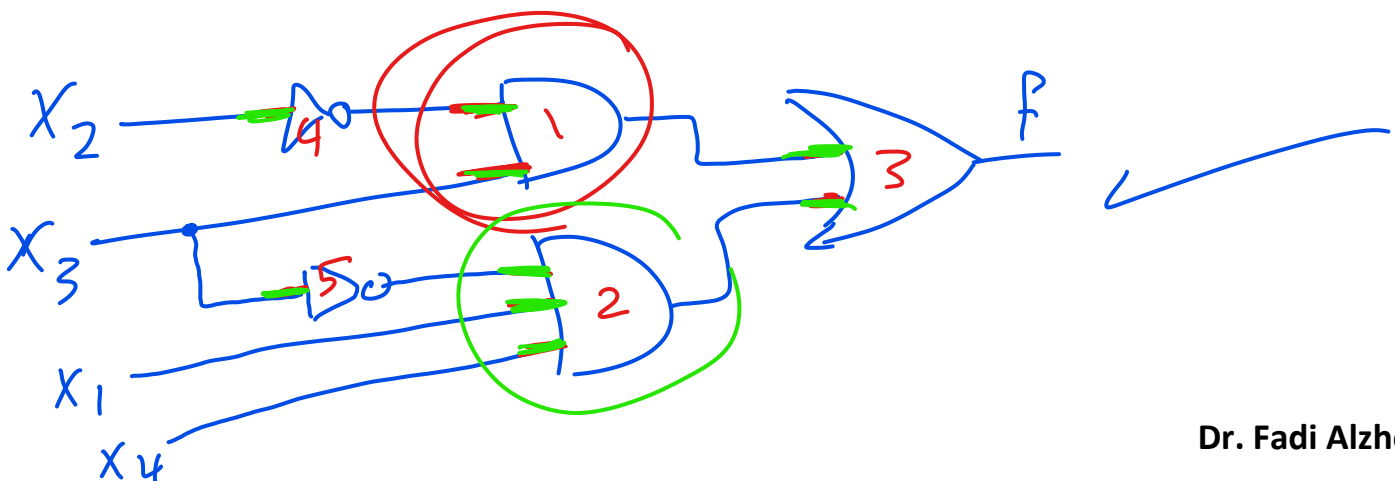
$x_1 x_2 \backslash x_3 x_4$	00	01	11	10
00	0 ⁰	0 ⁴	0 ¹²	0 ⁸
01	0 ¹	0 ⁵	1 ¹³	1 ⁹
11	1 ³	0 ⁷	0 ¹⁵	1 ¹¹
10	1 ²	0 ⁶	0 ¹⁴	1 ¹⁰

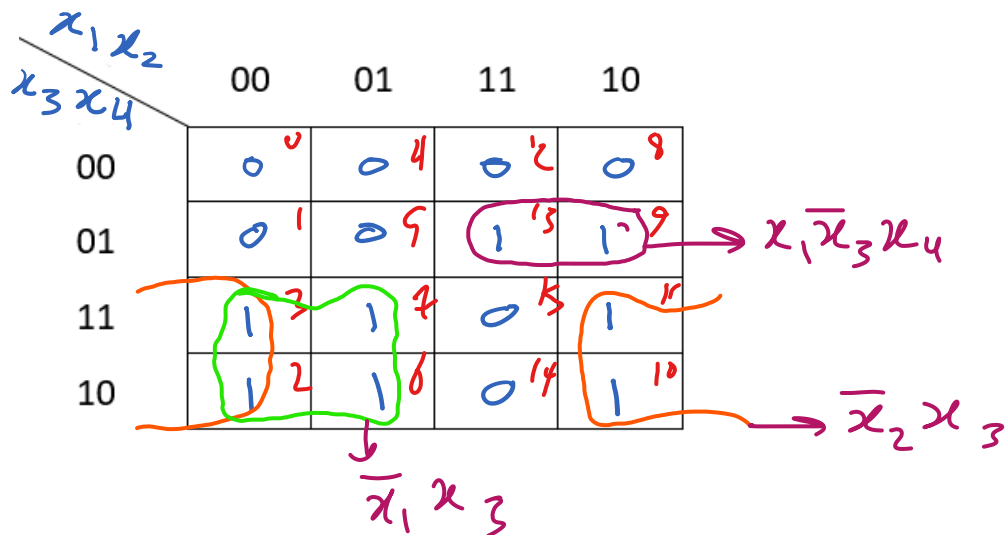
Handwritten annotations: $x_1 \bar{x}_2 x_3 x_4$ (circled 13, 9), $\bar{x}_2 x_3$ (circled 3, 11, 2, 10)

$$f = \bar{x}_2 x_3 + x_1 \bar{x}_3 x_4$$

$$\text{Cost of } f = 2 \text{ ANDs} + 1 \text{ OR} + 2 \text{ NOTs}$$

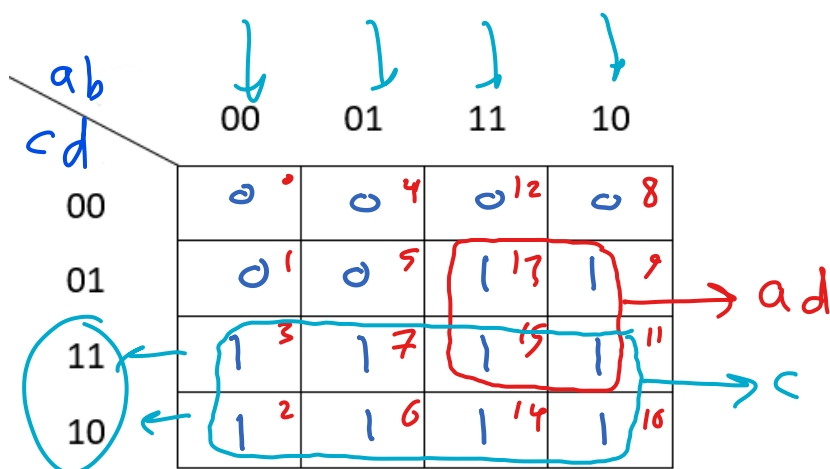
$$+ 9 = 14$$





$$f = \bar{x}_1 x_3 + \bar{x}_2 x_3 + x_1 \bar{x}_3 x_4$$

min cost
Sop



$$f = c + a d$$

Find min cost SOP

ab
cd

	00	01	11	10
00	1 ⁰	0 ⁴	0 ¹²	1 ⁸
01	0 ¹	0 ⁵	0 ¹³	0 ⁹
11	1 ³	1 ⁷	1 ¹⁵	0 ¹¹
10	1 ²	1 ⁶	0 ¹⁴	1 ¹⁰

$\bar{b}\bar{d}$

bcd

$\bar{a}c$

1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

$$f = \bar{a}c + bcd + \bar{b}\bar{d}$$

	00	01	11	10
00				
01				
11				
10				